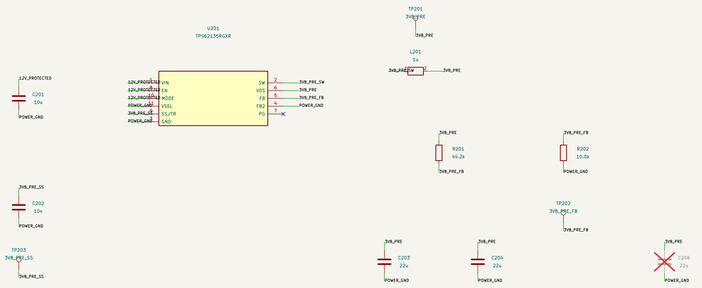
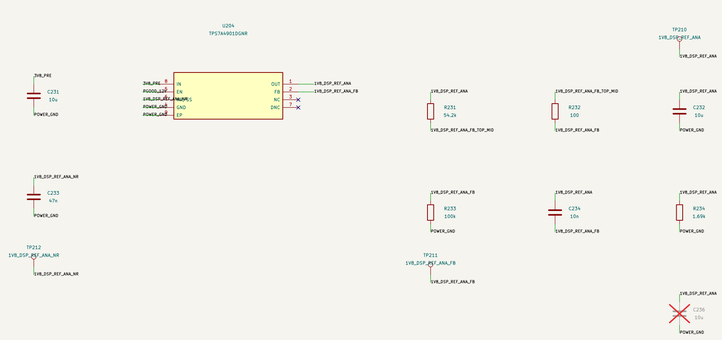
3V8_PRE | TPS62135 forced PWM



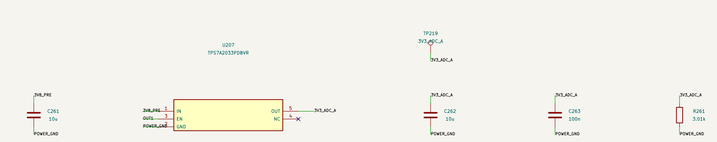
MODE=VIN: VSEL=0: FB2=0: native PG intentionally NC.
Kelvin FB bottom to GND; VOS to local COUT positive.
L/C qualification and distributed capacitance remain board validation.

1V8_DSP_REF_ANA | TPS7A4901DGNR



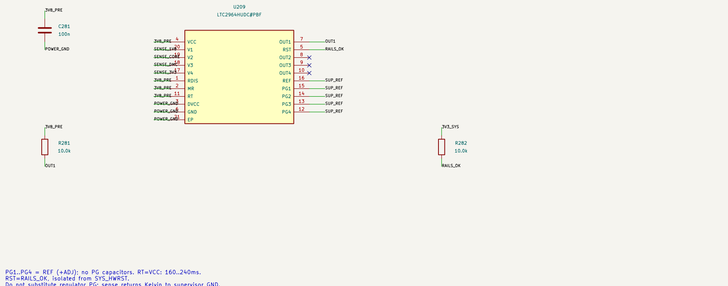
DNC pin 7 and NC pin 3 open; EP pin 9 grounded.
Preload is permanent. Included in total rail allocation.
CFF spans OUT to FB (including series top-arm resistors).

3V3_ADC_A | TPS7A2033PDBVR



P variant: active output discharge. EN=OUT1.
No startup-time guarantee extrapolated to this capacitor bank.
Consumer-side filtering/decoupling belongs to the later load sheets.

Four DSP rail comparators / first-rail qualification



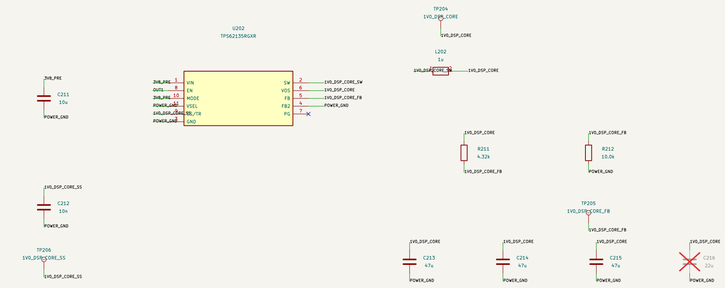
PG1..PG4 = REF (+AD): no PG capacitors. RT=VCC: 160..240ms.
RST=RAILS_OK, isolated from SYS_HWRST.
Do not substitute regulator PG: sense returns Kelvin to supervisor GND.

PHASE 4B BLOCKED: common OUT1 cannot turn microphones off while retaining core/DMC power.
Phase 3D section 9.1(5), 11(6) versus section 9.2 commanded-shutdown order. No enable relation changed.

AFE EN held off by R310. Command and fault override NOT connected.

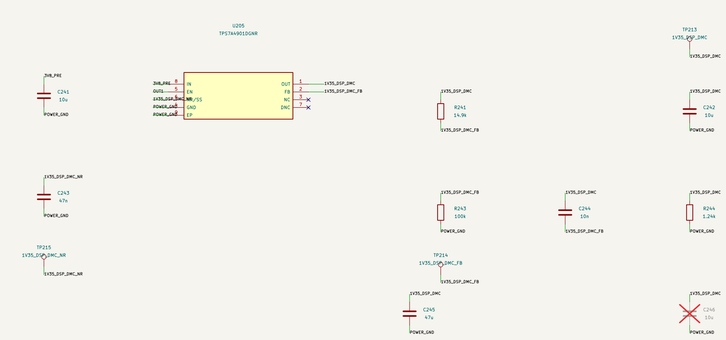
No power-up / reset / safe-state implementation approval is implied.
Hardware-safe latch, dominance, commanded shutdown, and AFE override are incomplete.

1V0_DSP_CORE | TPS62135 forced PWM



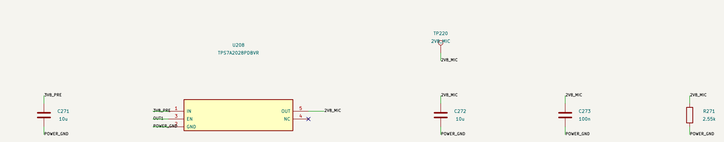
MODE=VIN: VSEL=0: FB2=0: native PG intentionally NC.
Kelvin FB bottom to GND; VOS to local COUT positive.
L/C qualification and distributed capacitance remain board validation.

1V35_DSP_DMC | TPS7A4901DGNR



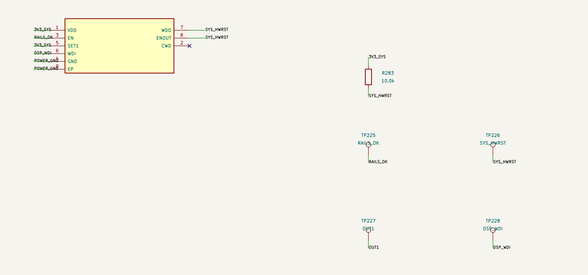
DNC pin 7 and NC pin 3 open; EP pin 9 grounded.
Preload is permanent. Included in total rail allocation.
CFF spans OUT to FB (including series top-arm resistors).

2V8_MIC | TPS7A2028PDBVR



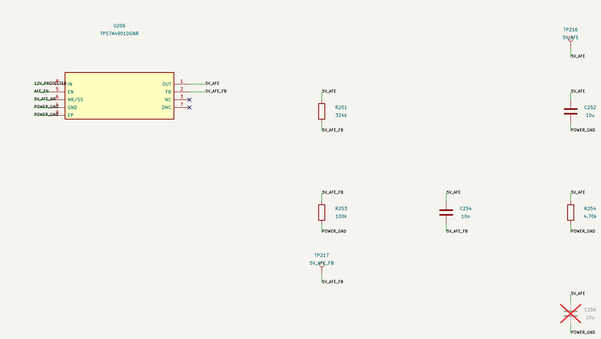
P variant: active output discharge. EN=OUT1.
No startup-time guarantee extrapolated to this capacitor bank.
Consumer-side filtering/decoupling belongs to the later load sheets.

TPS3431: independent watchdog / delayed reset



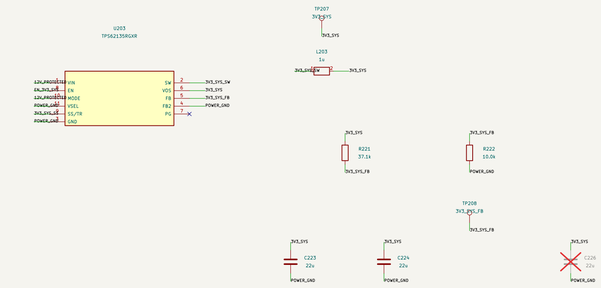
CWD NC; SET1=VDD: WDO and ENOUT wired open drain.
350..470ms total rail-valid to reset release; WDT 1.36..1.84s.
First DSP failing WDI edge <=1.13s after reset release.
DSP WDI default low; firmware services during boot.

5V_AFE | TPS7A4901DGNR



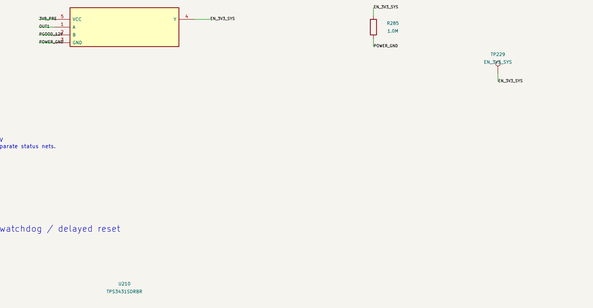
DNC pin 7 and NC pin 3 open; EP pin 9 grounded.
Preload is permanent. Included in total rail allocation.
CFF spans OUT to FB (including series top-arm resistors).

3V3_SYS | TPS62135 forced PWM



MODE=VIN: VSEL=0: FB2=0: native PG intentionally NC.
Kelvin FB bottom to GND; VOS to local COUT positive.
L/C qualification and distributed capacitance remain board validation.

5V_AFE | TPS7A4901DGNR



DNC pin 7 and NC pin 3 open; EP pin 9 grounded.
Preload is permanent. Included in total rail allocation.
CFF spans OUT to FB (including series top-arm resistors).

3V3_SYS EN = OUT1 AND PGOOD_12V

1.0M EN pulldown; inputs remain separate status nets.

TPS3431: independent watchdog / delayed reset

CWD NC; SET1=VDD: WDO and ENOUT wired open drain.
350..470ms total rail-valid to reset release; WDT 1.36..1.84s.
First DSP failing WDI edge <=1.13s after reset release.
DSP WDI default low; firmware services during boot.

AFE EN held off by R310. Command and fault override NOT connected.
No power-up / reset / safe-state implementation approval is implied.
Hardware-safe latch, dominance, commanded shutdown, and AFE override are incomplete.

Partial entry only. Commanded-shutdown enable conflict; hardware-safe path incomplete.

circuitBuilderAI-Astra

Sheet: /
File: power_regulation.kicad_sch

Title: 2 — Power Regulation / Sequencing

Size: User	Date: 2026-09-13	**Rev: 4B-BLOCKED**
KiCad E.D.A. 10.0.6		Id: 1/1